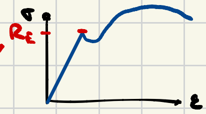
Frage 1 ~ Haie



Spannung vs Streckgrenze

$$\sigma = \frac{F}{A} = \frac{2\,70\,\text{kg} \cdot 9.81\,\frac{m}{s^2}}{0.25\,\text{cm}^2} \approx 54\,\text{MPa}$$

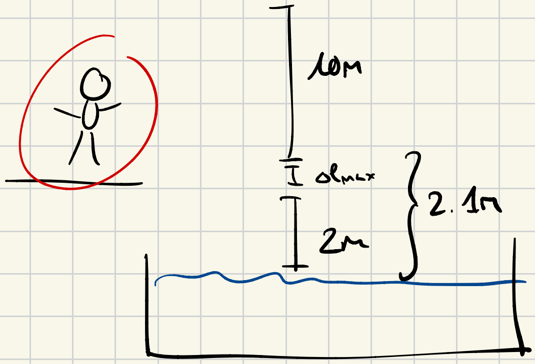
$$E = \frac{\sigma}{\varepsilon} \quad ; \quad \varepsilon = \frac{\Delta l}{l}$$

↳ $\Delta l = \frac{2F \cdot L}{A \cdot E}$ ⇒

a) $\Delta l = 700\,\text{mm}$

b) $\Delta l = 11.2\,\text{mm}$

c) $\Delta l = 200\,\text{mm}$



B

Frage 3 ~ Polybahn

$F_{max} = 40\,\text{kN}$

$S = 3$

$R_e = 1100\,\text{MPa}$ → $\frac{F}{R_e} \cdot S = A$

$A = ?$ ↳ same same für anderen

Frage 10 ~ Lennard-Jones

$$U_{LJ}(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

$\varepsilon = 0.26\,\text{eV}$ $\quad \sigma = 2.6\,\text{Å}$ $\quad (1\,\text{Å} = 10^{-10}\,\text{m})$

Was ist r_{Glgw}? (in Å!)

$$4\varepsilon\left(-12\frac{\sigma^{12}}{r^{13}} \ldots\right)$$

↳ nochmals ableiten & $=0$ setzen

$1\,\text{eV} = 1.6 \cdot 10^{-19}\,\text{J}$; $1\,\text{Å} = 10^{-10}\,\text{m}$